Hints :

1. Since gravitational force is a central force, therefore net torque on the planet is zero. Hence angular momentum is conserved and areal velocity is directly proportional to angular momentum. $\therefore$ Law of areas is a consequence of conservation of angular momentum.

2. Since earth-satellite system is a bound system, therefore total energy is negative.

3. Satellite is accelerated towards the centre of earth. It is in the state of free fall.

4. Moon has no atmosphere because escape velocity on its surface is less than RMS speed of molecules of gas. Hence gas molecules, if any, will escape from it.

5. Since gravitational force is a central force,
$\therefore$ Angular momentum and mechanical energy is conserved
Since force on particle is non-zero
$\therefore$ Linear momentum is not conserved

6. Since escape speed = $\sqrt{\dfrac{2GM}{R+h}}$, it do not depend on mass of body.

7. Geostationary satellite rotates from west to east, must be in equitorial plane
and is at a height of 35800 km from surface of earth.

8. $PE = -\dfrac{GMm}{r}, \; KE = \dfrac{GMm}{2r}$

If $r\downarrow \Rightarrow PE\downarrow$ व $KE\uparrow$

9. $K = \dfrac{4\pi^2}{GM}$

where M is mass of sun. It is same for all planets.

10.

$\vec{F} = m\vec{I}$	$U = mV$
$MLT^{-2} = M[I]$	$ML^2T^{-2} = M[V]$
$\Rightarrow [I] = LT^{-2}$	$[V] = L^2T^{-2}$

11. Galileo did experiments with bodies rolling down inclined planes to recognize the fact that all bodies are accelerated equally towards earth's surface.

12. In geocentric model, all celestial objects moved around earth in circular orbits.

13. Ptolemy proposed geocentric model 2000 years ago.

14. Heliocentric model in which planets revolved around the sun was metioned by Aryabhatta (5^{th} century A.D.) in his treatise.

17. It is the property of ellipse

18. Central force $\Rightarrow \vec{\tau} = \vec{0} \quad \Rightarrow \vec{L}$ = constant

$\Rightarrow \dfrac{d\vec{A}}{dt} = \dfrac{\vec{L}}{2m} = \text{constant}$

20. Yes they can have the same speed. The orbital speed is independent of the mass of the satellite.

21. The value of acceleration due to gravity is less on hills

22. An astronaut experiences weightlessness in a space satellite. Its is because the astronaut experiences no gravity.

23. $g' = g - \omega^2 R_e \cos^2\lambda$

24. The gravitational potential energy is maximum at infinite.

25. Because gravitational potential energy arises due to attractive force